



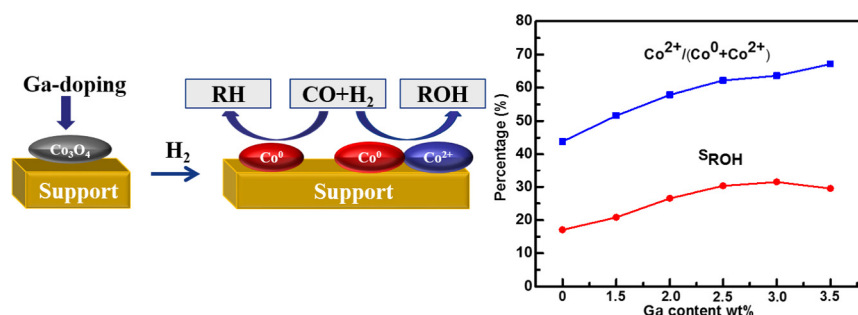
Effects of gallium as an additive on activated carbon-supported cobalt catalysts for the synthesis of higher alcohols from syngas

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GRAPHICAL ABSTRACT



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ABSTRACT

Direct synthesis of higher alcohols by CO hydrogenation is one of the most important processes for the utilization of syngas, but obtaining high selectivity to higher alcohols (C₂₊–OH) remains a challenge. In this work, a series of gallium-doped activated carbon-supported cobalt catalysts (15Co–xGa/AC, representing 15 wt% Co and x wt% Ga) were developed and they were found to be favorable for enhancing the selectivity to alcohols (S_{ROH}). The optimized S_{ROH} value reached 30.3 mol% over the 15Co–2.5Ga/AC catalyst, while that was 17.0 mol% over the Ga-free catalyst. The characterization results indicated that the doping of Ga increased the reduction temperature of Co₃O₄ and modulated the distribution of the surface Co²⁺ and Co⁰ species. It was found that the Co²⁺–Co⁰ dual active sites were responsible for the formation of alcohols and the enhancement of S_{ROH} arose from moderate ratio of Co²⁺/(Co²⁺ + Co⁰). The results revealed that the addition of Ga could be an effective way to elevate the selectivity of higher alcohols by CO hydrogenation.

1. Introduction

The term higher alcohols generally refers to alcohols with carbon number above two, which have been considered a chemical feedstock for the production of fuels, fuel additives, plasticizers, detergents, lubricants, and so on [1–4]. Currently, the production of higher alcohols

in large-scale industrial process is performed through homogeneous catalysis [5,6]. The production involves two processes: hydroformylation of an olefin into an aldehyde and subsequent hydrogenation to the respective alcohol. Unlike homogeneous catalysis, heterogeneous catalysis can synthesize alcohols in one step via the direct conversion of syngas over solid catalysts. To date, four types of catalysts have been

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extensively studied for the direct synthesis of higher alcohols (HAs) from syngas, namely, modified methanol catalysts [7–9], modified Fischer–Tropsch synthesis (FTS) catalysts [10–12], Mo-based catalysts [13–15], and Rh-based catalysts [16–18]. Although substantial progress has been made for the synthesis of HAs, the process still faces major challenges in improving the activity, alcohol selectivity, and long-term stability [19].

In recent decades, Co-based catalysts has been extensively investigated for its excellent ability of carbon chain growth. The selectivity of C_2+ -OH products, especially C_6+ -OH, over cobalt-based catalysts is superior to the traditional modified Fischer–Tropsch CoCu and CuFe bimetallic catalysts [20–26]. Several mechanisms have been proposed for the formation of alcohols by cobalt-based catalysts. Baker et al. [27] and Blanchard et al. [28] proposed that Co^{2+} - Co^0 dual active sites were responsible for the synthesis of alcohols in their explanation for the production of a significant amount of alcohols over silica- and alumina-supported cobalt catalysts in CO hydrogenation. Ding group [5,20,21] found that the formation of Co_2C during the reaction was favorable for the production of alcohols and that a moderate ratio of Co_2C/Co was beneficial for the improvement of S_{ROH} . Density functional theory calculation [5] suggested that Co_2C (111) could serve as the active site for non-dissociative CO adsorption. Thus, the authors concluded that Co- Co_2C dual active sites were essential for the production of alcohols. Gnanamani et al. [29,30] claimed that the diffusion of active hydroxyl groups from the Co–Ce junction to reactive C_xH_y species resulted in the formation of oxygenates over a Pt–Co/CeO₂ catalyst.

However, majority of alkanes and negligible oxygenates are formed over the traditional Co-based catalysts by CO hydrogenation. Hence, some appropriate promoters are introduced to improve the selectivity of alcohols over cobalt-base catalysts. For instance, it was reported that the doping of La₂O₃ enhanced the formation of alcohols and even elevated the CO conversion at a moderate loading [20]. From characterization of the spent catalysts, the authors assigned the enhanced selectivity for alcohols to the formation of Co_2C . The doping of CaO had a similar promotion effect on S_{ROH} [21]. Unfortunately, the doping of CaO hindered the dispersion of cobalt and resulted in a decrease in CO conversion. A similar promotion effect on S_{ROH} was found for the doping of Al₂O₃ [31], K [32], and Li [33] by the same group. In addition, it was recently demonstrated that Ce as a promoter had a positive effect on the production of alcohols over a 15Co5.0Ce–CNT catalyst [34]. The authors attributed the promotion effect to the formation of a Co–CeO_{2-x} site, and the mechanism of alcohols synthesis was similar to that proposed for the Pt–Co/CeO₂ catalyst [29,30].

The incorporation of gallium (Ga) into catalysts for the synthesis of HAs was first investigated in a CoGaZn–Al₂O₃ system by Ning and co-workers [35]. The authors constructed CoGaZn–Al₂O₃ layered double hydroxides that became layered double oxides after calcination. The CoGaZn–Al₂O₃ catalyst had a uniformly dispersed and trapped structure, which displayed an excellent catalytic performance for the synthesis of HAs. The alcohols selectivity over the optimized composition catalyst reached 59.0 mol% at a CO conversion of 43.5 mol%. The authors lately proposed that the synergism between Co species in CoGa alloys and isolated Co sites was responsible for the synthesis of alcohols [36]. However, the role of Ga as an additive on activated carbon supported cobalt catalyst prepared by traditional impregnation method and studies on the effect of Ga have been scarce up to now.

In this work, we prepared a series of Ga-doped activated carbon-supported Co catalysts for the synthesis of HAs from syngas. The detailed structural study and the discussion of relationship between catalyst structure and catalytic performance addressing the effects of Ga incorporation were shown. A dual active site of Co^{2+} - Co^0 was then proposed to account for the formation of alcohols by CO hydrogenation.

2. Experimental

2.1. Catalyst preparation

The supported Co-based catalysts were prepared by a conventional incipient wetness co-impregnation method. Before preparation of the catalysts, the activated carbon support (AC, Tianjin Calgon Carbon Co. Ltd) was washed with hot deionized water several times to remove impurities according to the procedure reported in literature [5]. The detailed procedure was as follows: raw AC that originated from coconut shells was added into deionized water and stirred at 100 °C for 0.5 h, then filtered and washed several times with deionized water. The obtained AC was added into water again, and the water washing procedure was repeated 10 times, after which the AC was dried at 100 °C for 12 h. The catalysts were synthesized by impregnating the AC with a mixed aqueous solution of $Co(NO_3)_2$ and $Ga(NO_3)_3$. After impregnation, the fresh catalysts were dried at 100 °C for 12 h and calcined at 350 °C for 5 h in an Ar atmosphere. The cobalt loading was 15.0 wt% and the Ga doping content was x wt% ($x = 1.5, 2.0, 2.5, 3.0$, and 3.5). The obtained catalysts were labelled 15Co- x Ga/AC.

2.2. Catalyst testing

Evaluation of the catalytic performance was carried out in a micro fixed-bed reactor (reactor tube 9.0 mm i.d.). A thermocouple was put into the reactor and placed in contact with the catalyst to monitor the catalyst bed temperature. In general, to avoid hot spots, 0.3 g catalyst was mixed with 1.7 g of quartz sand and placed in the middle of the reactor tube. Prior to reaction, the catalyst was reduced at 450 °C for 5 h with pure H₂. After the reactor cooled to the designated temperature, premixed syngas was purged through the reactor for 2 h to remove residual H₂. Then, the reactor was slowly pressurized to 3.0 MPa. The composition of syngas was $H_2/CO/N_2 = 6/3/1$ molar ratio, where N₂ was used as an internal standard. The gaseous and liquid products were analyzed on-line by gas chromatography. The outlet stream was passed through a cold trap. The gases in the stream, such as CO, N₂, CO₂ and CH₄, were analyzed by a GC (GC9790, Zhejiang Fuli Analytical Instrument Co., Ltd.) equipped with a thermal conductivity detector (TCD) and a TDX-01 packed column; the other hydrocarbons and oxygenates were analyzed by an Agilent 7890 GC equipped with a flame ionization detector and a KB-PONA capillary column. The GC chart indicated that the column was capable of analyzing the alcohols and olefins completely as shown in Fig. S1. All data were collected at steady state after 20 h of reaction.

2.3. Catalyst characterization

X-ray diffraction (XRD) was conducted on a Rigaku Ultima IV diffractometer using Cu K α radiation ($\lambda = 0.15406$ nm) with a scanning angle from 20° to 70° at a speed of 10°/min, a tube voltage of 40 kV and a current of 40 mA. The phase composition of the catalyst was determined by matching the obtained diffraction data with that in the JCPDS database.

Temperature programmed reduction (TPR) was performed with a Micromeritics AutoChem II 2920 chemisorption analyzer. Approximately 0.1 g of the catalyst sample was first pretreated in a flow of Ar at 150 °C for 0.5 h to eliminate moisture and then cooled to 50 °C. TPR was performed with 10% H₂-90% Ar from 50 °C to 800 °C at temperature ramp rate of 10 °C/min. The signal for the consumption of H₂ was collected by TCD. CuO was used to as a reference to calibrate the hydrogen consumption.

H₂ chemisorption measurement was done on Micromeritics ASAP 2020 surface area and porosity analyzer. The weight of the measured sample was approximately 0.20 g. The sample was first reduced for 5 h with pure H₂ at 450 °C in a U-type tube. After reduction, the sample was evacuated at that temperature for 2 h to eliminate the residual H₂. The

H₂-adsorption isotherms were carried out at 150 °C, as the temperature recommended by Reuel and Bartholomew [37]. The total H₂ uptake was achieved by extrapolating the linear part of the isotherm to zero pressure. The dispersion of cobalt was calculated according to the amount of hydrogen adsorption and a stoichiometry of one hydrogen atom per cobalt atom.

Transmission electron microscopy (TEM) was performed on a FEI Tecnai F30 electron microscope operated at an acceleration voltage of 300 kV. The catalysts were first reduced at 450 °C for 5 h under an atmosphere of H₂ and then ultrasonically dispersed in ethanol for 15 min. The obtained solution was dropped on carbon-coated copper grids.

X-ray photoelectron spectroscopy (XPS) was performed using an Omicron Sphera II photoelectron spectrometer equipped with an Al K_α X-ray radiation source ($h\nu = 1486.6$ eV). Prior to measurement, each sample was pressed into a thin disk and reduced at 450 °C for 5 h under an atmosphere of H₂ in the in-situ cell. After reduction, the sample was introduced into the XPS chamber. All operations were performed under vacuum to avoid exposure to air. The binding energy was calibrated using the C 1s peak at 284.6 eV.

3. Results and discussion

3.1. Catalytic performance

Table 1 shows the catalytic performance results over all 15Co-*x*Ga/AC catalysts. The catalysts were evaluated for the synthesis of HAs by CO hydrogenation at 220 °C, 3.0 MPa and a H₂/CO molar ratio of 2 to determine the effect of Ga on the catalytic activity and product selectivity. Hydrocarbons and alcohols were simultaneously produced on all 15Co-*x*Ga/AC catalysts with minor CO₂ formation. On the 15Co/AC catalyst, the conversion was 25.9 mol%. After modification with 1.5 wt % Ga, the conversion increased slightly to 28.9 mol%. Upon further increasing the amount of Ga, the conversion decreased to the minimum value of 5.2 mol% over 15Co-3.5 Ga/AC. This decrease in conversion suggested that the addition of Ga had some adverse effects on CO conversion.

In terms of the selectivity for alcohols over the 15Co-*x*Ga/AC catalysts, Table 1 shows that the *S*_{ROH} was 17.0 mol% over the 15Co/AC catalyst, and the alcohols were mainly C₂₊-OH with a C₂₊-OH/ROH ratio of 89%. It should be noted that the ratio of C₂₊-OH/ROH over the 15Co/AC catalyst was higher than those of the traditional CuCo [22,25,26] and CuFe catalysts [23,24]. As shown in Table 1, the *S*_{ROH} value improved with increasing Ga content from 20.8 mol% over 15Co-1.5Ga/AC to the maximum value of 31.4 mol% over 15Co-3.0Ga/AC. However, the ratio of C₂₊-OH/ROH decreased to 85%–80% upon incorporation of Ga. These results suggested that the doping of Ga enhanced the formation of alcohols but decreased the proportion of C₂₊-OH in the total alcohols. This effect was similar to that observed for Ca-promoted Co-Ca/AC catalysts [21]. In addition, it should be mentioned that, different with the remarkable alcohols selectivity achieved by 15Co-*x*Ga/AC, common silica- and alumina-

Table 1
The catalytic performance over 15Co-*x*Ga/AC catalysts.

Catalyst	Conv. (mol%)	Selec. (mol%)	Distribution of alcohols (mol%)				
			ROH	RH	CO ₂	CH ₃ OH	C ₂₊₅ -OH
15Co/AC	25.9	17.0	81.5	0.5	11.0	43.4	45.6
15Co-1.5Ga/AC	28.9	20.8	78.4	0.8	15.7	48.9	35.4
15Co-2.0Ga/AC	21.8	26.5	73.0	0.5	19.8	40.3	39.9
15Co-2.5Ga/AC	13.1	30.3	68.7	1.0	19.2	56.3	24.5
15Co-3.0Ga/AC	10.6	31.4	66.4	2.2	15.6	58.4	26.0
15Co-3.5Ga/AC	5.2	29.5	67.8	2.7	18.4	63.1	18.5

Reaction conditions: 220 °C, 3.0 MPa, GHSV = 4000 mL/(g·h).

Table 2

The comparison of CO conversion and alcohols selectivity reported in literatures and this work.

Catalyst	Reaction conditions	CO conv. (mol%)	Selec. to alcohols (mol %)		Refs.
			Total alcohols	C ₂₊ -OH	
15Co/AC	3.0 MPa, 900 h ⁻¹ , 220 °C	64.3	15.4	10.2	[21]
15Co-0.1Ca/AC		49.0	30.6	27.0	
15Co/AC	3.0 MPa, 500 h ⁻¹ , 220 °C	65.2	20.1	19.6	[31]
15Co-0.2Al ₂ O ₃ /AC		68.5	27.7	26.6	
15Co/CNTs	2.5 MPa, 10000 mL/ (g·h), 270 °C	78.8	12.3	4.9	[34]
15Co-5.0Ce/CNTs		30.6	38.6	23.3	
15Co/AC	3.0 MPa, 4000 mL/ (g·h), 220 °C	25.9	17.0	15.1	This work
15Co-2.5Ga/AC		13.1	30.3	24.5	

supported cobalt catalysts produce negligible alcohols, and the value of *S*_{ROH} is usually below 10 mol% [21,22]. It may be due to support and promoter effect. For comparison, representative carbon material supported cobalt-based catalysts were listed in Table 2. Taking the activity, selectivity and reaction conditions into considering, 15Co-*x*Ga/AC will be one of the excellent catalysts for the synthesis of HAs.

The selectivity to a product is generally related to the level of conversion. Low conversion is usually promoted to obtain a high selectivity for certain product and vice versa. For the 15Co-2.5Ga/AC catalyst, the *S*_{ROH} value was 30.3 mol%, which was the second highest value among the investigated catalysts. However, it should be noted that the high *S*_{ROH} value was achieved under the condition of low conversion of 13.1 mol%. Low conversion might lead to a high *S*_{ROH} value at some degree. Therefore, to investigate the effect of Ga on *S*_{ROH}, it is necessary to eliminate the impact of conversion on *S*_{ROH}. By adjusting the space velocity, the comparison of *S*_{ROH} over the 15Co-2.5Ga/AC and 15Co/AC catalysts was performed under similar conversions. Table 3 shows the catalytic results of the 15Co-2.5Ga/AC and 15Co/AC catalysts under space velocities of 2000 mL/(g·h) and 4000 mL/(g·h), respectively. As shown in Table 3, the conversions over the two catalysts were similar, but the *S*_{ROH} values were still different. Although the *S*_{ROH} value of 15Co-2.5Ga/AC decreased from 30.3 mol% at a conversion of 13.1 mol% to 26.5 mol% at a conversion of 28.6 mol%, the value was still higher than that of 15Co/AC at similar conversions. This finding suggested that the introduction of Ga was indeed beneficial for the formation of alcohols.

3.2. Catalyst characterization

Fig. 1 displays the XRD patterns of the as-calcined catalysts. As shown in Fig. 1, the characteristic peaks of Co₃O₄ were detected as the only phase with 2θ at 36.8°, 44.7°, 59.3° in all samples. The diffraction peaks of Co₃O₄ became broadened relative to the pristine counterpart after the addition of Ga into the host 15Co/AC catalyst. This broadening suggested that the addition of Ga to the host catalyst could enhance the dispersion of Co in the range investigated.

Fig. 2 shows the TPR profiles of 15Co-*x*Ga/AC. There were three reduction peaks in the TPR profiles for all the examined catalysts. As for 15Co/AC sample, the first two peaks were attributed to the typical two-step reduction of Co₃O₄. The former was due to reduction of Co₃O₄ to CoO, and the latter was ascribed to reduction of CoO to Co [38,39]. The broad peak located at 400–600 °C was ascribed to gasification of the AC support [20,21]. However, as for the Ga doping catalysts, especially with high Ga content samples, the reduction of Ga species could not be neglected and displayed some reduction peaks in TPR spectrum. To confirm the assignment of Ga species reduction peak, the TPR of 3 Ga/

Table 3
The catalytic performance over 15Co-xGa/AC catalysts at different space velocities.

Catalyst	Conv. (mol%)	Selec. (mol%)		Distribution of alcohols (mol%)			
		ROH	RH	CO ₂	CH ₃ OH	C ₂₋₅ -OH	C ₆₊ -OH
15Co/AC ^a	25.9	17.0	81.5	0.5	11.0	43.4	45.6
15Co-2.5Ga/AC ^b	28.6	26.5	73.0	0.5	17.0	59.6	23.4

^a Reaction conditions: 220 °C, 3.0 MPa, GHSV = 4000 mL/(g·h).

^b Reaction conditions: 220 °C, 3.0 MPa, GHSV = 2000 mL/(g·h).

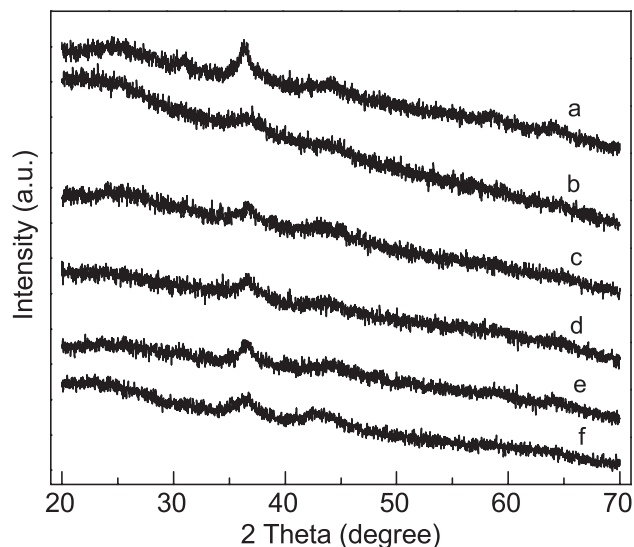


Fig. 1. XRD patterns of the 15Co-xGa/AC catalysts: (a) 15Co/AC, (b) 15Co-1.5Ga/AC, (c) 15Co-2.0Ga/AC, (d) 15Co-2.5Ga/AC, (e) 15Co-3.0Ga/AC and (f) 15Co-3.5Ga/AC.

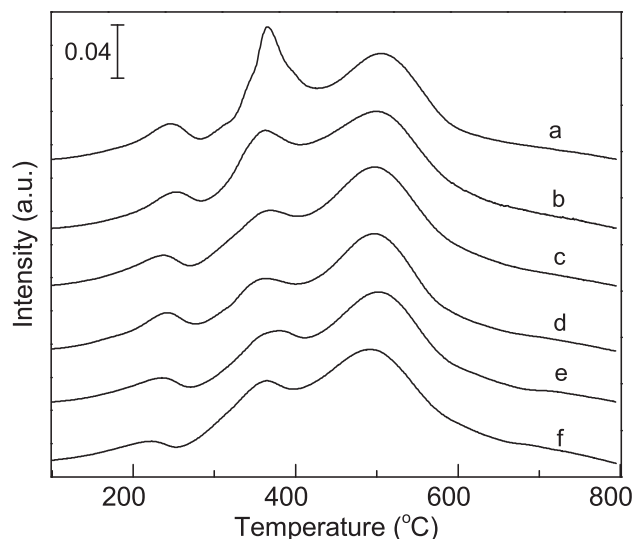


Fig. 2. H₂-TPR profiles of the 15Co-xGa/AC catalysts: (a) 15Co/AC, (b) 15Co-1.5Ga/AC, (c) 15Co-2.0Ga/AC, (d) 15Co-2.5Ga/AC, (e) 15Co-3.0Ga/AC and (f) 15Co-3.5Ga/AC.

AC and AC were performed and shown in Fig. S2. There were two reduction peaks in the TPR profile of 3Ga/AC which were attributed to the reduction of Ga₂O₃ located at 400–500 °C [40,41] and gasification of AC support at high temperature [20,21]. Therefore, the second reduction peak in the TPR of 15Co-xGa/AC was attributed to the reduction of Ga₂O₃ and CoO to Co. In addition, it can be seen from Fig. 2 that the intensity of the second reduction peak obviously decreased

Table 4

H₂ uptake, reduction and dispersion degree, the ratio of Co²⁺/(Co⁰ + Co²⁺) for 15Co-xGa/AC catalysts.

Catalyst	H ₂ uptake ^a (μmol/g _{cat})	Reduction degree ^b (%)	Dispersion degree ^c (%)	Co ²⁺ /(Co ⁰ + Co ²⁺) ^d (%)
15Co/AC	52.4	52.8	4.1	43.8
15Co-1.5Ga/AC	78.6	41.1	6.2	51.6
15Co-2.0Ga/AC	73.4	34.5	5.7	57.9
15Co-2.5Ga/AC	62.9	27.7	4.9	62.2
15Co-3.0Ga/AC	57.6	25.3	4.5	63.6
15Co-3.5Ga/AC	47.8	24.6	3.7	67.1

^{a,c}Based on H₂ chemisorption results, assuming one hydrogen atom absorbed on one cobalt surface atom.

^bThe reduction degree based on the results of Co₃O₄ reduction peaks in the TPR profiles.

^dBased on corresponding peak areas in XPS.

after the addition of Ga. This result meant that the doping of Ga hindered the reduction of CoO to Co, resulting in a higher content of CoO species in the reduced catalysts. It could suggest a strong interaction existing between Ga and Co species. Considering the partial overlap of reduction peak between CoO and Ga₂O₃, the degree of reduction was calculated by integrating and fitting the area of the Co₃O₄ reduction peak in the TPR profiles and then subtracting the peak area of Ga₂O₃. The results are presented in Table 4. The degree of reduction decreased from 52.8% over 15Co/AC to a minimum value of 24.6% over 15Co-3.5Ga/AC. This result further suggested that the addition of Ga hindered the reduction of Co₃O₄, and especially that of CoO to Co.

The amount of surface Co sites was measured by H₂ chemisorption, and the results are summarized in Table 4. It can be observed from Table 4 that the degree of Co dispersion in the 15Co-xGa/AC catalysts decreased from 6.2% to 3.7% when the Ga doping content increased from 1.5 wt% to 3.5 wt%. Therefore, it could be concluded that the introduction of a moderate amount of Ga was beneficial for enhancing the dispersion of Co.

To identify the Co species and analyze their distributions in the reduced catalysts, XPS characterization was conducted (Fig. 3A). The Co 3p XPS spectra of the reduced catalysts displayed two main peaks were located at 780.0 eV and 795.0 eV, corresponding to Co 2p_{3/2} and Co 2p_{1/2}, respectively. The two main peaks were broad and asymmetric, indicating that there were different Co species on the surface of the reduced catalysts. The fitting results are presented in Fig. 3A. The Co 2p_{3/2} peak could be divided into Co⁰, Co²⁺, and satellites peaks. Among these peaks, Co⁰ 2p_{3/2} and 2p_{1/2} were located at 778.3 eV and 793.3 eV and those of Co²⁺ appeared at 780.4 eV and 795.6 eV, respectively [42,43]. The satellite peaks were centered at 784.5 eV and 801.7 eV. These results suggested the coexistence of Co⁰ and Co²⁺. From Fig. 3A, it could be observed that the addition of Ga enhanced the intensity of the Co²⁺ peaks but weakened those of Co⁰, which implied that the addition of Ga significantly affected the ratio of Co²⁺/(Co²⁺ + Co⁰). For quantitative analysis, the ratio of Co²⁺/(Co²⁺ + Co⁰) in the reduced samples was calculated by deconvolution of the XPS peaks, and the results are summarized in Table 4. As shown in Table 4, the

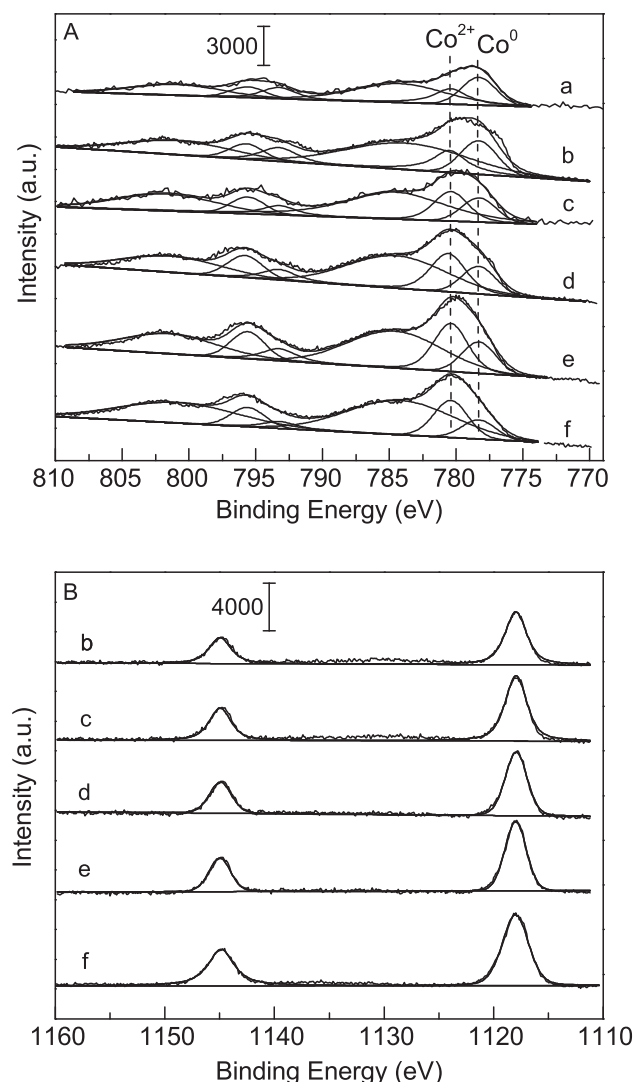


Fig. 3. XPS profiles of the 15Co-*x*Ga/AC catalysts (A) Co 2p XPS; (B) Ga 2p XPS. (a) 15Co/AC, (b) 15Co-1.5Ga/AC, (c) 15Co-2.0Ga/AC, (d) 15Co-2.5Ga/AC, (e) 15Co-3.0Ga/AC and (f) 15Co-3.5Ga/AC.

incorporation of Ga in the host 15Co/AC catalyst was beneficial for enhancing the ratio of $\text{Co}^{2+}/(\text{Co}^{2+} + \text{Co}^0)$; the ratio increased from 43.8% over 15Co/AC to the maximum value of 67.1% over 15Co-3.5Ga/AC.

Fig. 3B depicts the XPS profiles of Ga 2p for all the reduced samples. It displayed two sharp and symmetry peaks located at 1118.0 eV and 1145.0 eV corresponding to $2p_{3/2}$ and $2p_{1/2}$ which was ascribed to Ga_2O_3 according to the previous literature reports [44,45]. Hence, Ga_2O_3 hardly reduced at 450 °C for 5 h. But, it was worth noting that there were no changes in the binding energy of Ga and Co after the addition of Ga, which mean no electron transfer between Ga species and Co species. The result was different from the report by An et al. [36], in which the author found the formation of CoGa particle and existing the electron transfer from Ga sites to Co sites according to the result of XPS over CoGaZn- Al_2O_3 layered double hydroxides catalysts. In addition, An et al. also mentioned the formation of CoGa alloys in their catalyst system [36]. In order to confirm whether or not the formation of CoGa alloys in the present system, the XRD characterization for the reduced catalysts was conducted and the result is shown in Fig. S3. There are no reflection peak shifts after the addition of Ga. Combined with the aforementioned results of XRD and XPS, it can be concluded that there are no CoGa alloys formed in 15Co-*x*Ga/AC. The

finding is also contrary with the result by An et al. These may be essentially resulted from the different catalyst compositions and preparation conditions.

To examine the detailed crystalline structure of the particles in the reduced catalysts, HRTEM characterization was performed (Fig. 4). Well-resolved lattice fringes can be seen in the images in Fig. 4. The lattice spacing were measured to be approximately 0.15 nm, 0.21 nm and 0.24 nm, which corresponded to CoO (2 2 0) CoO (2 0 0) and CoO (1 1 1) planes, respectively. The interval of 0.20 nm and 0.18 nm were correspond to Co (2 0 0) and Co (1 1 1) planes. This result indicated that CoO was present in all samples after reduction in hydrogen. This finding was consistent with the results of the TPR and XPS characterizations.

3.3. Structure-performance correlation

In this study, we prepared catalysts by the introduction of a moderate amount of Ga into a pristine Co/AC catalyst, which was favorable for improving the Co dispersion and enhancing the formation of alcohols. The TPR spectra of the 15Co-*x*Ga/AC catalysts showed that the doping of Ga retarded the second-step reduction of Co_3O_4 , namely, the reduction of CoO to Co^0 . This result suggested that there were more Co^{2+} species in the reduced Ga-doped catalysts compared with the Ga-free one. Characterization by XPS revealed that the surface Co^0 and Co^{2+} distributions were significantly influenced by the addition of Ga. The ratio of $\text{Co}^{2+}/(\text{Co}^{2+} + \text{Co}^0)$ increased from 43.8% to 67.1% after incorporation of 3.5 wt% Ga into the 15Co/AC catalyst. In addition, the HRTEM images of reduced catalysts further verified the presence of CoO. Therefore, it could be concluded that the Co species existed in the form of CoO and metallic Co in the reduced catalysts; the ratio of CoO clearly increased after introduction of Ga into the Co/AC host catalyst.

To interpret the structure-catalytic performance relationship, the dependence of the S_{ROH} value on the ratio of $\text{Co}^{2+}/(\text{Co}^{2+} + \text{Co}^0)$ for the 15Co-*x*Ga/AC catalysts was examined, and the results are shown in Fig. 5. It can be seen from Fig. 5 that S_{ROH} increased with increasing ratio of $\text{Co}^{2+}/(\text{Co}^{2+} + \text{Co}^0)$. The result implies that a moderate ratio of $\text{Co}^{2+}/(\text{Co}^{2+} + \text{Co}^0)$ is beneficial for the production of alcohols and the highest S_{ROH} value is obtained on the 15Co-*x*Ga/AC catalysts with Ga doping at 2.5–3.0 wt%. Furthermore, previous studies have reported that the basicity is a pivotal condition for the synthesis of alcohols from CO hydrogenation, being in favor of CO chemisorption [24,46]. Thus the CO_2 -TPD and NH_3 -TPD measurements were specially performed and the results are displayed in Figs. S4 and S5. No obvious CO_2 desorption peak over Ga_2O_3 and no apparent changes in the shape of CO_2 desorption peaks over 15Co/AC and 15Co-2.5Ga/AC catalysts can be identified. The amount of CO_2 uptake over 15Co/AC and 15Co-2.5Ga/AC is estimated by integrating the CO_2 desorption peaks, approximately being 45.3 $\mu\text{mol/g}$ and 47.3 $\mu\text{mol/g}$, respectively. The negligible difference between them suggests that the addition of Ga does not significantly change the strength of basicity on the surface of catalysts. The NH_3 -TPD profile is displayed Fig. S5. No obvious NH_3 desorption peak can be observed, that is, the addition of Ga could not give rise to the change of acidity. Therefore, the CO_2 -TPD and NH_3 -TPD results indicate that the addition of Ga does not bring about the change of catalyst surface acidity or basicity. That means the promotion effect of Ga on the synthesis of alcohols is not via the change of acidity or basicity. Nevertheless, we cannot exclude the influences of acidity/basicity on the formation of higher alcohols by the results obtained with present catalyst systems. Based on the aforementioned characterization results, the promoter effects of Ga could be attributed to the increased ratio of CoO in the reduced catalysts after the addition of Ga to the host Co/AC catalyst. Consequently, Ga seemed to be a potential promoter for improving the selectivity to alcohols in the synthesis of HAs.

The synthesis of alcohols is well-known to need the synergistic effects of dual active sites, which are responsible for CO associative adsorption and CO dissociative adsorption. It is generally accepted that

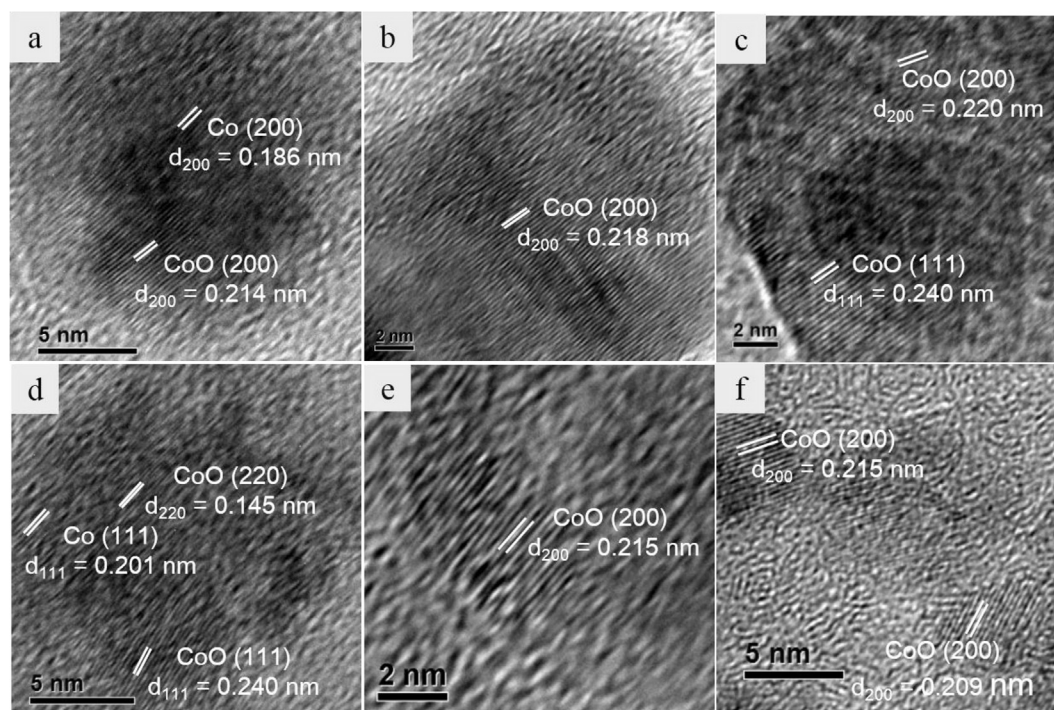


Fig. 4. HRTEM images of the 15Co-*x*Ga/AC catalysts: (a) 15Co/AC, (b) 15Co-1.5Ga/AC, (c) 15Co-2.0Ga/AC, (d) 15Co-2.5Ga/AC, (e) 15Co-3.0Ga/AC and (f) 15Co-3.5Ga/AC.

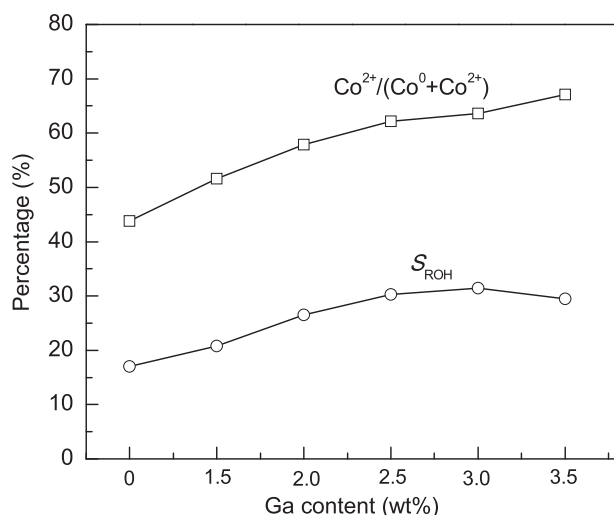


Fig. 5. Relationship between the ratio of $\text{Co}^{2+}/(\text{Co}^0 + \text{Co}^{2+})$ and the selectivity of alcohols over the 15Co-*x*Ga/AC catalysts with different Ga dopings.

the Co^0 species is the active site for CO dissociative adsorption. The dissociative CO adsorption step is followed by hydrogenation to form $^*\text{CH}_x$ species on the surface of metallic Co. The coupling of these $^*\text{CH}_x$ species leads to carbon chain growth, and long-chain hydrocarbons and alcohols are eventually formed [47,48]. The function of Co^{2+} species as the CO associative adsorption site has been demonstrated in many reports [27,28], and its presence favors the formation of alcohols [46,49]. For example, Baker et al. [27] and Blanchard et al. [28] found the production of a significant amount of higher oxygenates in CO hydrogenation using silica- and alumina-supported Co catalysts, and they both proposed that unreduced cobalt ions were responsible for the products. Smith et al. [49] claimed that Co^{2+} favored the formation of alcohols in studies on the CO adsorption behavior of Cu/SiO₂, Co/SiO₂ and CuCo/SiO₂ catalysts by in situ infrared spectroscopy. Ishida et al. proposed that the formation of Co^{2+} was conducive to improving the

selectivity for alcohols in the investigation of the effects of alkali metal promoters on Co-supported silica for the synthesis of HAs [46]. In addition, a similar perspective was recently proposed by Jiao [20] and Wang [22]. However, it has been reported that the synergic effect between Co_2C and Co metal is essential to the formation of alcohols [5,20,21]. Hence, it is necessary to confirm whether or not the Co_2C formed. At first, we conducted the XRD characterization of spent catalysts. As shown in Fig. S6, no diffraction lines were assigned to Co_2C but Co metal. The results suggest no bulk Co_2C phase was formed during the reaction. We then conducted the C 1s XPS characterization for the spent catalysts. As shown in Fig. S7 in the Supplementary Material of revised manuscript, the binding energies at 284.6 eV, 286.0 eV and 289.0 eV are seen in the C 1s XPS profiles for the spent catalysts 15Co/AC and 15Co-2.5Ga/AC; these peaks are assigned to graphite carbon, C–O groups and COOH groups on the catalyst surfaces [50,51], respectively. According to literature [51,52], the C 1s peak of Co_2C is appeared at 283.6 eV. Therefore, combined with the results of XRD and XPS, we conclude that there is no Co_2C formed on the present catalyst surfaces during the reaction. The similar result was obtained by An et al. in their investigation on the layered double hydroxides-derived catalysts CoGaZn- Al_2O_3 for the synthesis of HAs [36]. Nonetheless, An et al. proposed that the dual active site Co–CoGa alloy was related to the synthesis of HAs [36]. In contrary to this, we consider that the species of Co and Ga are two isolated phases but closely contacted with each other on the surfaces of 15Co-*x*Ga/AC catalysts according to the results of various characterizations such as XPS, TEM and XRD. Due to this strong interaction, part of Co species are restrainedly reduced into Co^0 , resulting in a higher concentration of Co^{2+} species in the Ga–Co/AC catalysts and thereby a beneficial effect for the synthesis of HAs. More specifically, by excluding the formation of Co_2C species and CoGa alloys as the active sites, we propose that the dual active sites of Co^0 – Co^{2+} might be responsible for the synthesis of HAs in our system as illustrated in Fig. 6.

It is known that the conversion of syngas is a surface polymerization reaction. There are two different perspectives with respect to the rate determining step proposed according to the different chain growth reactions of CO hydrogenation, that is, the carbide chain growth

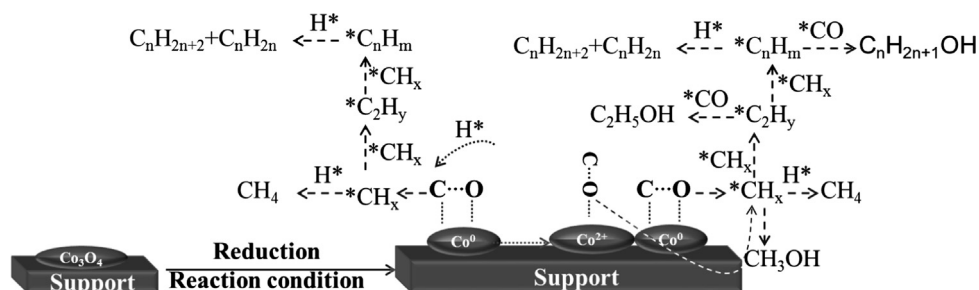


Fig. 6. Proposed reaction mechanism for the synthesis of HAs on Co⁰–Co²⁺ dual active sites.

mechanism and the CO insertion chain growth mechanism as shown in Fig. S8 [53,54]. In the case of carbide mechanism, the formation of CH_x monomeric from adsorbed C and H is generally considered as rate controlling [53,54]. In the mechanism of CO insertion, the rate of CO dissociation is slow compared with the rate of CO insertion and the rate of chain growth termination. Thus, the rate of CO dissociation is the rate determining step. It should be noted that the two mechanisms are proposed for FTS but not for the synthesis of HAs by CO hydrogenation. However, considering the similarity between FTS and the synthesis of HAs, it is reasonable to extrapolate the mechanism of FTS to the synthesis of HAs. As for the present work, due to the complexities of catalyst structure and the mechanism of CO hydrogenation, the above two chain growth mechanisms coupled with the corresponding two rate-determining steps should coexist in the reaction.

However, it is unfortunate that the doping of Ga is unfavorable for the conversion of CO. This phenomenon was related to the reduction of the amount of available metallic Co active sites after the addition of Ga. For Co-based catalysts, the extent of conversion in CO hydrogenation is decided by the amount of surface Co active sites. However, the amount of surface Co active sites is related to the dispersion and reduction of Co. Although the doping of Ga is favorable for enhancing the dispersion of Co and is thus beneficial for providing more surface Co active sites, the doping also hinders the reduction of Co at the same time, which is generally expected to reduce the amount of surface Co active sites. Therefore, the combination of the positive effect of Ga on the dispersion of Co and the negative effect of Ga on the reduction of Co leads to the observed variations in conversion over the 15Co–xGa/AC catalysts.

4. Conclusions

The appropriate incorporation of Ga into 15Co/AC significantly impacted the catalytic activity and alcohol selectivity for the direct synthesis of HAs from syngas. Although the addition of Ga was unfavorable to the conversion of CO, it was beneficial for the formation of alcohols. The maximum *S*_{ROH} value for the Ga-promoted catalysts was almost two times higher than that of 15Co/AC.

A series of characterizations were performed to explain the relationship between the catalytic performance and the structure. The doping of Ga hindered the second-step reduction of Co₃O₄ and modulated the distribution of the surface Co species. The coexistence of Co²⁺ and Co⁰ was observed in the reduced samples. The dual active sites of Co⁰–Co²⁺ were proposed to be responsible for the formation of alcohols. Furthermore, a good relationship between the ratio of Co²⁺/(Co²⁺ + Co⁰) and *S*_{ROH} was established and the highest *S*_{ROH} value was obtained on the 15Co–xGa/AC catalysts with Ga doping at 2.5–3.0 wt%. Therefore, the moderate ratio of Co²⁺/(Co²⁺ + Co⁰) induced by the doping of Ga might be responsible for the enhanced *S*_{ROH} values.

Acknowledgement

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fuel.2018.05.048>.

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Supplementary Materials

Effects of gallium as an additive on activated carbon-supported cobalt catalysts for the synthesis of higher alcohols from syngas

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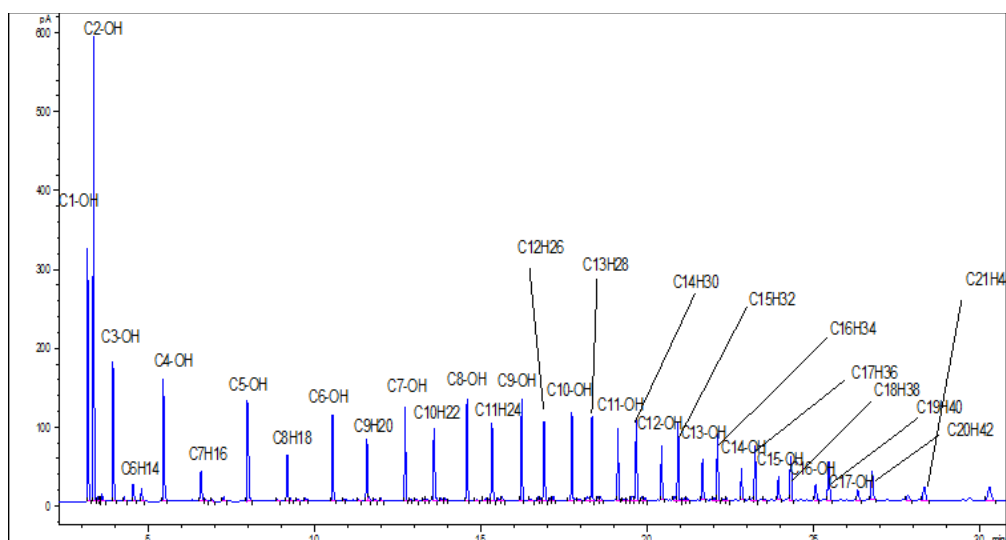


Fig. S1. Off-line GC chart of liquid products collected from cold trap over 15Co–2.0Ga/AC.

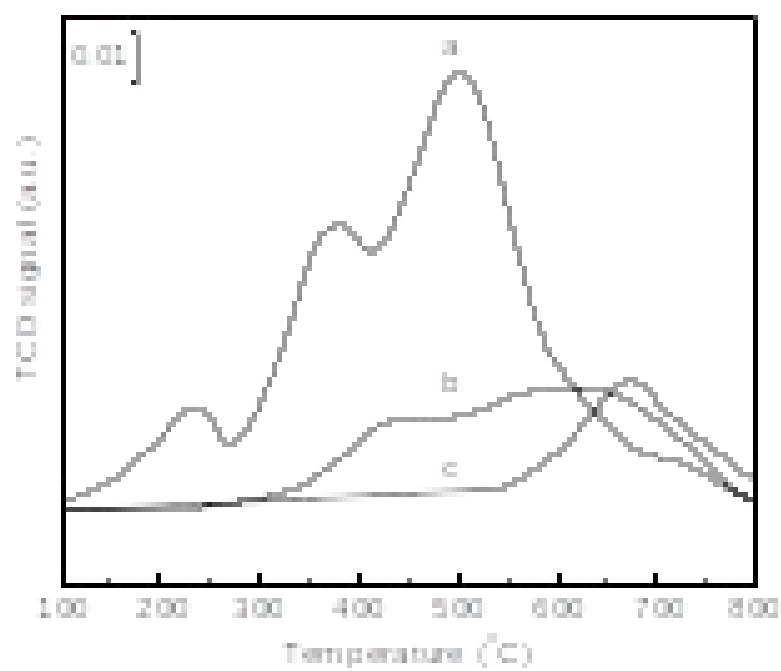


Fig. S2. TPR profiles of a: 15Co–3.0Ga/AC; b: 3.0Ga/AC; c: AC.

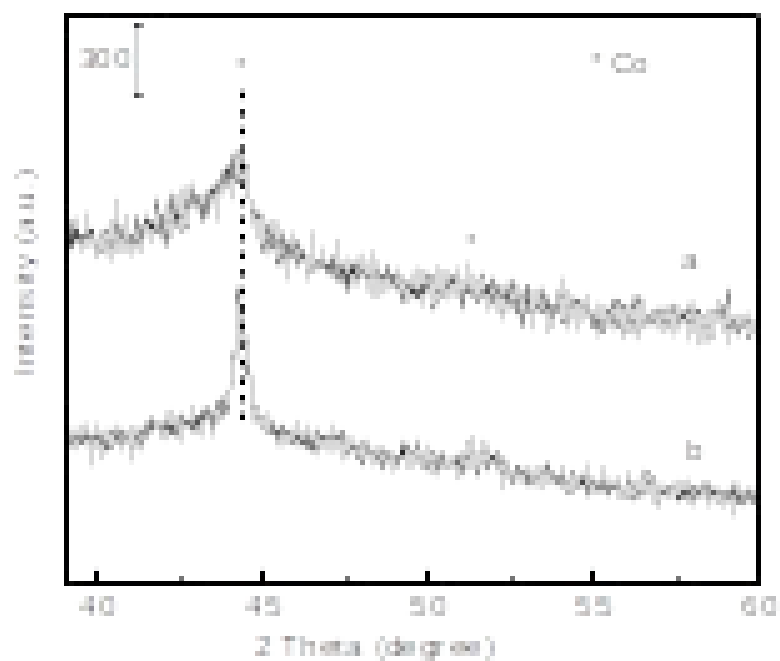


Fig. S3. XRD profiles for the reduced catalysts: a. 15Co/AC and b. 15Co–2.5Ga/AC.

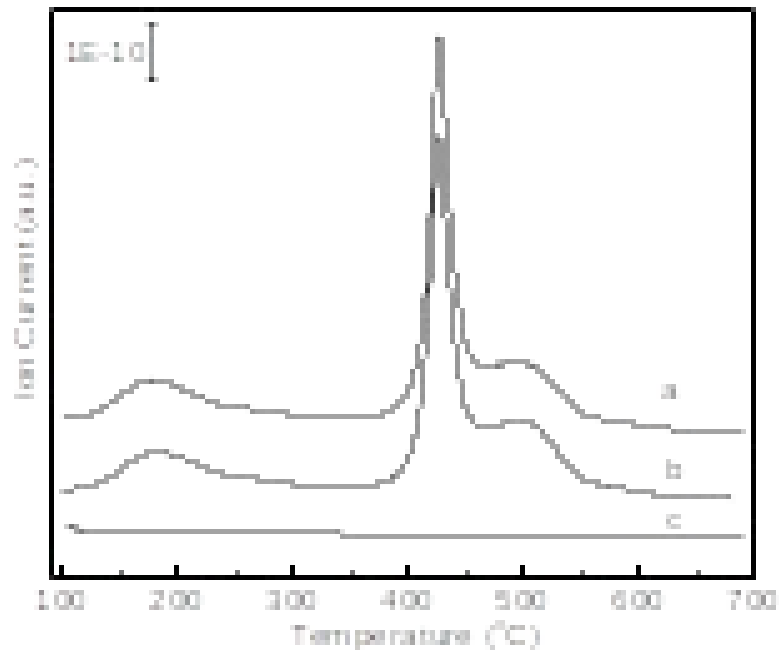


Fig. S4. CO₂-TPD profiles for the fresh catalysts: a. 15Co/AC; b. 15Co–2.5Ga/AC; c. Ga₂O₃.

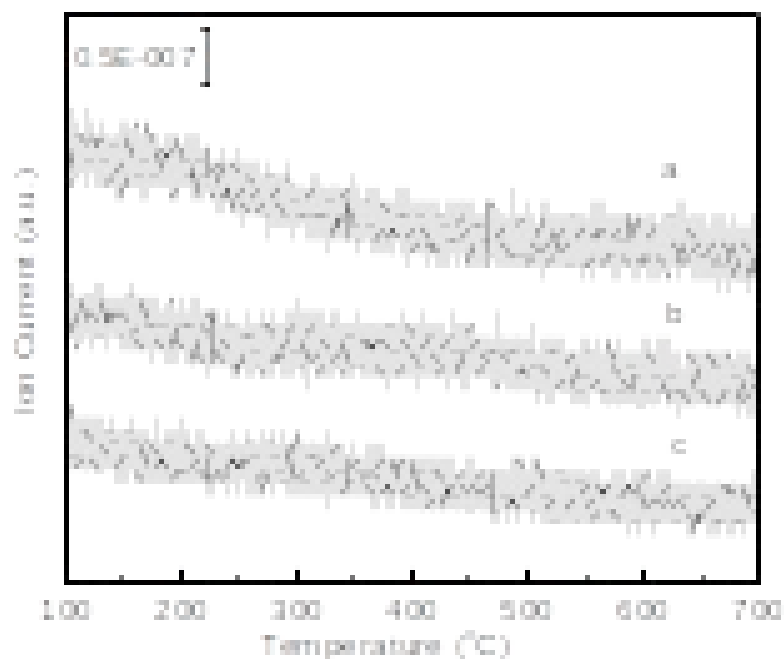


Fig. S5. NH₃-TPD profiles for the fresh catalysts: a. 15Co/AC; b. 15Co-2.5Ga/AC; c. Ga₂O₃.

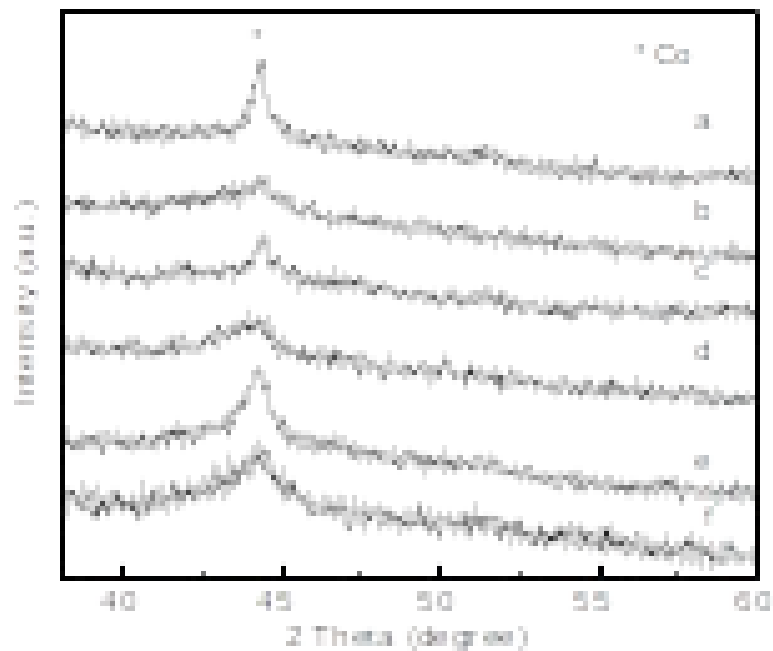


Fig. S6. XRD patterns of the 15Co-*x*Ga/AC catalysts after reaction: (a) 15Co/AC, (b) 15Co-1.5Ga/AC, (c) 15Co-2.0Ga/AC, (d) 15Co-2.5Ga/AC, (e) 15Co-3.0Ga/AC and (f) 15Co-3.5Ga/AC.

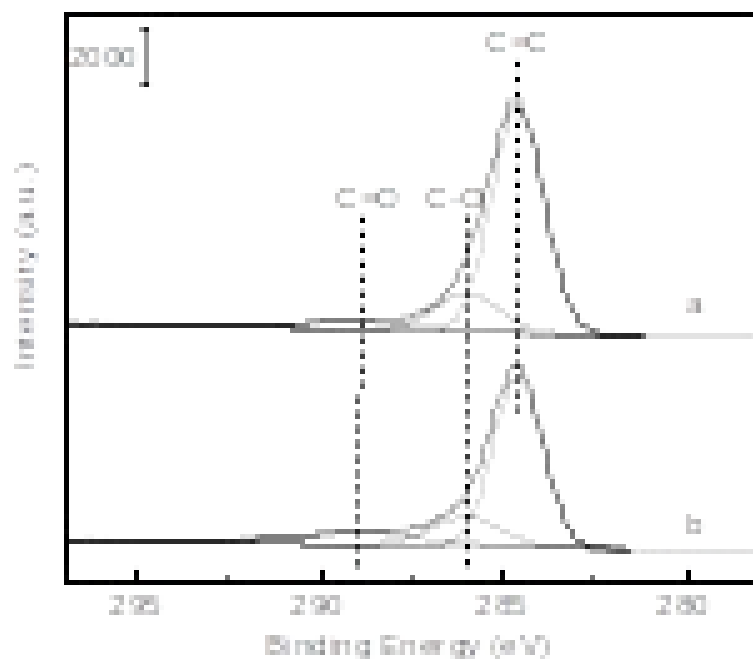


Fig. S7. XPS profiles of C 1s for the spent catalysts: a 15Co/AC and b 15Co–2.5Ga/AC.

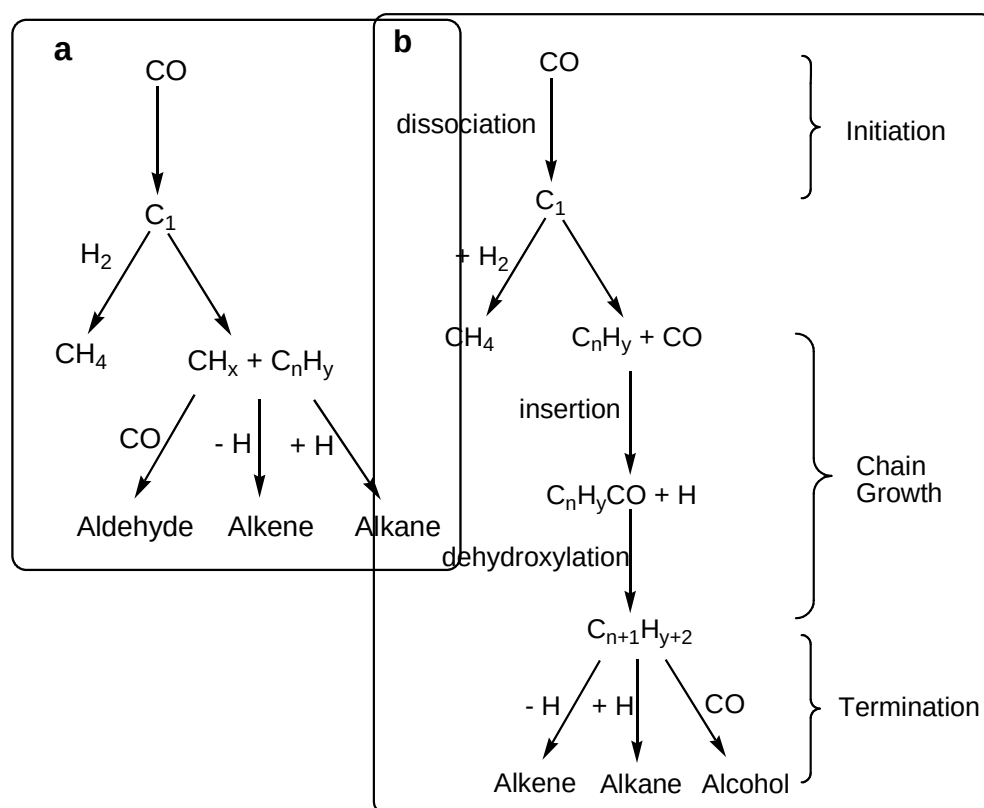


Fig. S8. Plausible reaction mechanism for CO hydrogenation: (a) carbide chain growth

mechanism and (b) CO insertion chain growth mechanism.